

The Decompression Cosmos: Information as Driver of Cosmic Expansion.

A Synthesis of Gravitational Information Theory and Informational Inevitability

David Carboni preprint

January 2026

Abstract

Cosmic expansion presents a foundational puzzle: how should we understand the relation between the universe's earliest state, the emergence of spacetime and mass-energy, and the lawful growth of structure across scales? This paper proposes an informational ontology in which information is fundamental and dynamically creative, while never arbitrary. Information unfolds within a pre-existing ontological grammar: the constraint architecture of reality that permits some relations, transformations, and forms while excluding others. Physical laws are stable, empirically legible expressions of this grammar, and energy and matter are downstream modes of instantiated informational structure.

Drawing on Vopson's information-theoretic treatment of gravity and the author's framework of informational inevitability, the paper develops the speculative hypothesis that early cosmic expansion may be understood as informational decompression: the progressive unfolding of a highly compressed informational state into spacetime, mass-energy, structure, life, and consciousness. The analogy is not to a grammarless file capable of becoming anything whatsoever. As in language, generativity is possible because expression is constrained. The cosmos possesses a wide but structured space of permitted unfolding.

This framework distinguishes global informational unfolding from local information optimization, proposes a constraint gradient in which additional restrictions accumulate through physical, chemical, biological, and cognitive instantiation, and clarifies the relation between representation and construction. Simulations, dreams, and mathematical models can represent states that cannot be materially built, not because they escape physical law, but because representing a possibility and constructing it are differently constrained activities.

The paper offers empirical research directions while explicitly separating established evidence, interpretive support, and speculative extension. Its central claim is ontological: the universe is not merely a collection of things that carry information, but a lawful informational unfolding whose creative dynamics operate within the grammar of reality.

Keywords: information cosmology, gravitational information theory, cosmic expansion, information decompression, Big Bang, consciousness emergence, Vopson, quantum mechanics, constraint gradient

1. The Mass-Lightness Problem

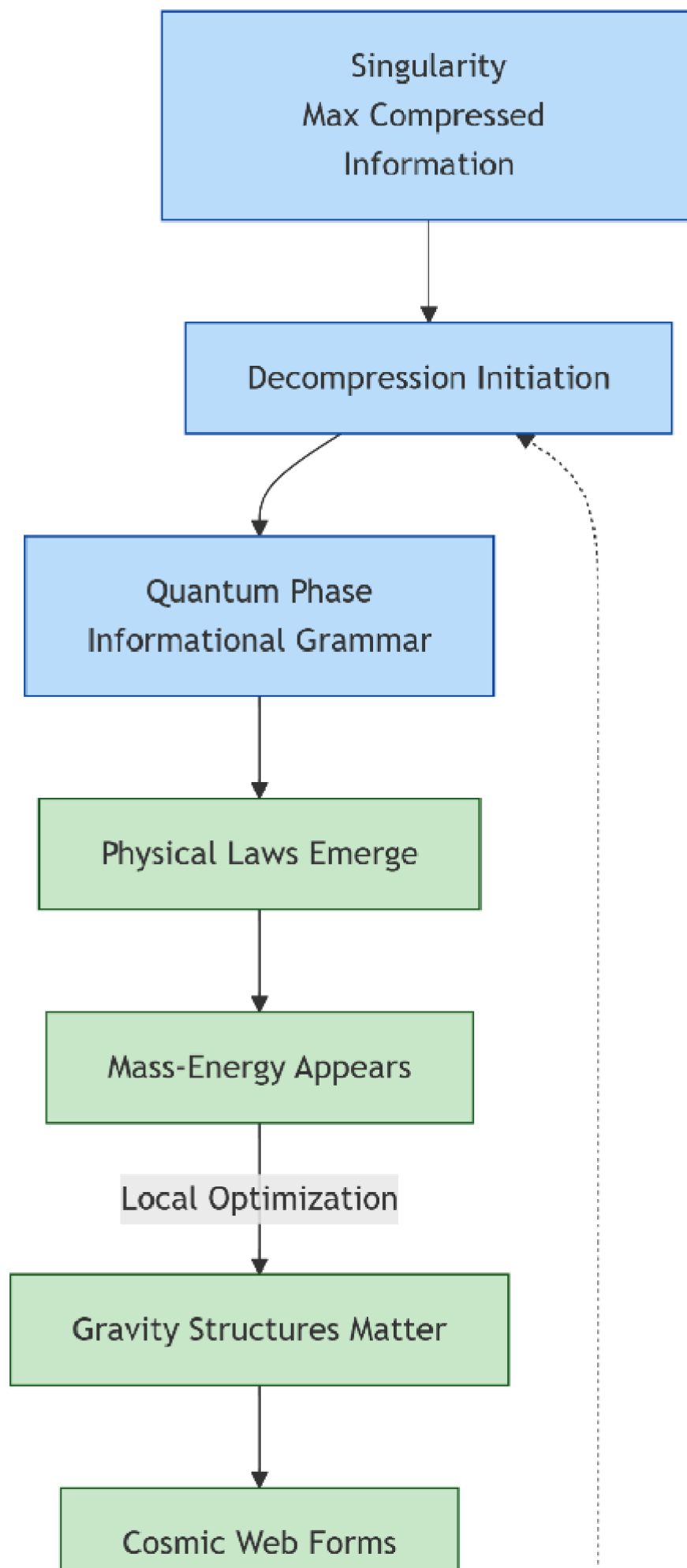
Standard Big Bang cosmology faces a fundamental tension. The universe began in a state of extreme density and expanded rapidly—that much we can observe. But the explanation creates problems: we attribute explosive expansion to mass-energy dynamics operating under infinite density conditions.

Here's the issue: mass resists rapid change. That's what inertia means. Mass is precisely the property that makes objects hard to accelerate. So how could mass-energy dynamics drive the fastest expansion event in cosmic history?

Something that needs to expand explosively should be:

Light (minimal inertial resistance)

Highly compressible (able to pack enormous content in tiny volume)



Capable of rapid decompression (no physical limits on unpacking speed)

Mass-energy fails on all counts. A primordial organizing principle—pure logic or rule set—succeeds on all counts.

This paper synthesizes three insights:

Vopson (2025): Gravity is information entropy reduction. Gravitational attraction is a computational optimization process that clusters matter to minimize information complexity. This means gravity is fundamentally informational, not purely physical.

Carboni (2025): The universe's initial conditions encoded an informational architecture that makes complexity inevitable. Cosmic structure, biological life, and consciousness aren't accidents—they follow from foundational informational constraints present from the beginning.

New insight: The initial singularity contained maximal compressed information (as algorithmic structure or logical rules) with minimal mass. The Big Bang is the decompression—the execution of this primordial logic—unleashing structured expansion. Energy and mass are the logical structure emerging from information. Its lightness is the kind of thing we would expect as an organizing principle in a cosmic pre-expansion state, before mass had increased exponentially due to physical interactions and engendered growth across scales.

The explanatory power: Information's relative lightness makes it the perfect driver of explosive expansion. Whatever caused cosmic inflation had to be unburdened by inertial mass. Information—as an organizing principle—fits perfectly. Mass-heavy structure is what information produces as it decompresses, not what drives the expansion.

2. Vopson's Information Gravity: The Foundation

Before we can argue that information drove cosmic expansion, we need to establish that information has physical reality. Vopson (2025) provides exactly this foundation.

2.1 Gravity as Information Entropy Reduction

Vopson demonstrates that gravitational attraction can be understood as an information optimization process:

Space consists of elementary cells (pixels) that store information about matter location.

When gravity clusters matter, it reduces the number of occupied cells.

Fewer occupied cells = lower information entropy.

Therefore: gravity is a computational process minimizing information entropy.

Mathematically: $I = k \log(N)$ where I is information content, N is the number of possible microstates, and k is a constant. When particles cluster gravitationally—moving from many cells to fewer cells— N decreases, so I decreases. Information entropy is reduced.

This is intended as more than a metaphor, but the evidential status must be stated carefully:

Vopson proposes an information-theoretic account of gravitational dynamics. The present paper treats that proposal as a suggestive foundation, not as settled proof that gravity is fundamentally informational. Gravity emerges from information optimization.

2.2 What This Means for Cosmology

If gravity is fundamentally informational, several implications follow:

The cosmic web is literally an information processing network. Galaxy clusters aren't just matter in space—they're information optimally organized.

Structure formation is information optimization. When galaxies form, the universe is finding efficient ways to store and process information about matter distribution.

Mass-energy is secondary to informational organization. Physical law emerges from informational constraints, not the other way around.

Physical constants and symmetries are parameters of information processing, not arbitrary features of "matter."

This inverts traditional ontology. We usually think information is an abstraction we use to describe physical systems. But what if physical systems are manifestations of informational dynamics? What if information—as primordial logic or organizing principle—is the fundamental substrate, and matter-energy is what information looks like when instantiated? That's the framework we're building.

3. The Singularity: Maximal Information, Minimal Mass

3.1 The Traditional Infinite Density Problem

Standard Big Bang cosmology says the universe began from a state of infinite density—all mass-energy compressed to a point. This creates multiple fatal problems:

Infinite density breaks general relativity. Einstein's equations don't work at singularities.

Quantum gravity becomes unavoidable but untestable. We need a theory we don't have.

Causal structure becomes undefined. Time and causation break down.

Initial conditions seem physically impossible. How do you get infinite density without infinite energy?

Physicists know this is a problem. Most assume "something else" must be happening at the Planck scale. But what?

3.2 Quantum Mechanics as Informational Grammar

To understand the singularity, we first need to understand quantum mechanics—the physics governing the earliest moments. Relational and information-centered interpretations associated with Rovelli, Zeilinger, Brukner, QBism, and quantum information theory motivate the possibility that some quantum puzzles can be reframed in terms of relations, information, and observer-dependent state assignments. These interpretations are contested and should not be presented as having established that quantum mechanics is nothing but information. Key reinterpretations:

Superposition: Not that particles are physically in multiple states simultaneously, but that we have incomplete information from our observational perspective. (Rovelli's Relational QM)

Wave function collapse: Not a physical process where something literally "collapses," but an information update when observers interact with systems. (Zeilinger)

Entanglement: Not instantaneous influence across space, but shared information established through past interactions. The "spooky action" is an artifact of thinking particles are independent when they're actually a unified informational entity. (Quantum Information Theory)

Measurement problem: Dissolves if quantum states represent observer-relative information rather than objective physical reality. There's no mystery about "how measurement causes collapse" because nothing collapses—information updates. (QBism, Brukner)

This "It from Bit" perspective (Wheeler, 1990) transforms quantum mechanics from a puzzle requiring explanation into the grammar of information itself—the rules governing how information relates, updates, and correlates within networks.

Implication for cosmology: If quantum mechanics describes informational relationships, the quantum era of the early universe wasn't about physical particles in uncertain states. It was about information in its most compressed, pre-instantiated form. The Planck epoch represents maximum information density before decompression into spacetime structure.

3.3 Information Density Without Mass Density

Here's our alternative to infinite mass density: The singularity was a state of maximal information density with minimal mass.

Information—as a primordial organizing principle—can be compressed to extreme density without requiring extreme mass. The "infinite density" problem is reframed as maximal information compression, not mass compression. Physical laws emerge as the decompression algorithm encoded in the compressed information. Quantum effects reflect information discretization, not mass quantization.

We remain agnostic about whether information content was truly infinite or merely maximal-finite. The key point: information density can approach extremes that mass density cannot, because an organizing principle lacks inertial resistance.

3.3.1 Clarification on the Mass of Information

It is worth noting that even if future physics were to assign a non-zero mass-equivalent to information—for instance through extensions of the Landauer principle or other speculative equivalences—such a mass would be vanishingly small (on the order of $m_{\text{bit}} \sim 10^{-38} \text{ kg}$ per bit at room temperature). This value is so far below the scale of inertial significance that it would not alter the central argument: information remains effectively weightless in the context of cosmic expansion.

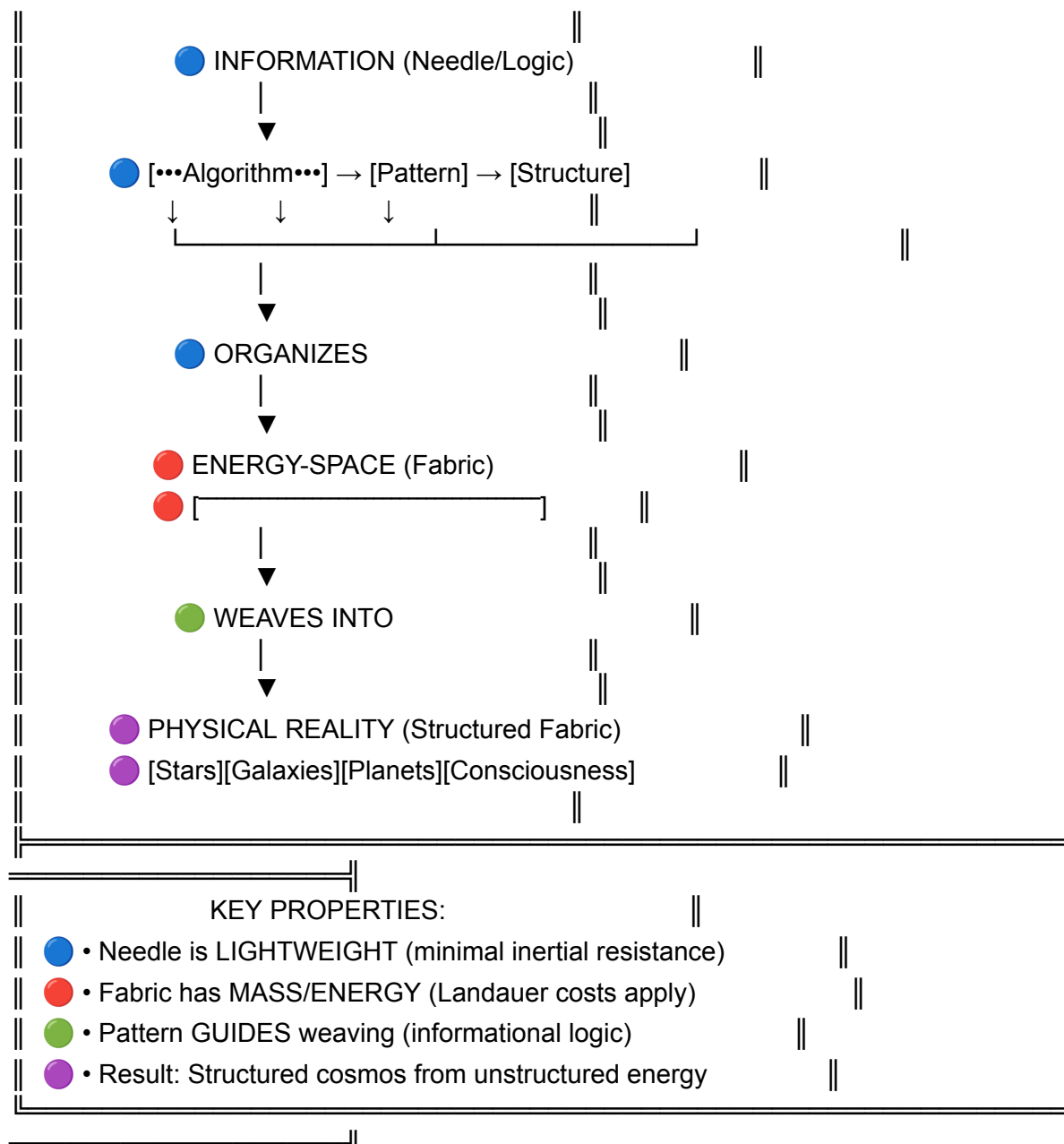
More importantly, such a minimal mass would not represent information as a “heavy” substance, but rather as the coupling mechanism through which logical structure interfaces with the physical substrate it organizes. Any residual mass would be a consequence of information’s instantiation—a trace of its binding to energy—not the source of its causal power. The “lightness” essential to our framework is therefore not contingent on literal masslessness, but on the ontological and functional primacy of information as an organizing principle. What drives expansion is not the negligible inertia of informational bits, but the explosive unpacking of compressed logical form—a process whose dynamics are governed by algorithmic necessity, not mechanical inertia.

Thus, whether information is massless or possesses a negligible coupling mass, its role remains unchanged: it is the logical precursor to mass-energy, not its competitor. The decompression cosmos is agnostic to this fine-grained physical detail; its explanatory power derives from the causal sequence itself—logic first, structure after.

3.4 Mechanistic Outline and Causal Sequence

The core mathematical challenge of this framework is to derive the equations of General Relativity (specifically the Friedmann equations governing expansion) from first principles of information decompression. We propose this would involve expressing the Hubble parameter $H(t)$ as a function of the information instantiation rate, dI/dt , where I represents the active, instantiated algorithmic information content of the cosmos. In this view, the cosmological constant (Λ) or dark energy may emerge as an integration constant of this informational dynamics, representing the residual momentum of decompression.





The causal sequence is:

Pre-Expansion State: Maximal compressed information (pure logic/algorithm/organizing principle).

Decompression Initiation: The logic begins to execute. This execution is expansion—light because it's the unfolding of a pattern, not the motion of stuff.

Mass Emerges Downstream: As the logic instantiates into relational structures (quantum fields, spacetime geometry, particle interactions), mass-energy appears as a property of those instantiated structures.

Exponential Growth of Massy Structures: Once sufficient mass-energy exists and is governed by the instantiated laws (gravity, nuclear forces), it can cluster, interact, and complexify—leading to the cosmic web, stars, planets, and life.

Energy and mass are the logical structure emerging from information. Its lightness is the kind of thing we would expect as an organizing principle in a cosmic pre-expansion before mass had increased exponentially due to physical interactions engendered growth across scales.

4. Big Bang as Information Decompression

4.1 Reframing Cosmic Inflation

Traditional inflationary theory: An unknown "inflaton field" causes exponential spatial expansion for a brief period, then decays into matter-energy. This explains observed large-scale structure but requires introducing an otherwise unobserved field with finely-tuned properties.

Information decompression view: The compressed informational state at the singularity unpacks according to its encoded algorithm. What we observe as spatial expansion is the manifestation of information decompression across dimensional axes.

Key advantages:

Natural mechanism for rapid expansion: Information decompression can be explosive because it has no inertial resistance. No need to posit special fields with mysterious properties.

Explains fine-tuning: Physical constants aren't arbitrary values requiring anthropic explanation—they're parameters of the decompression algorithm, inherent in the compressed information structure.

Resolves horizon problem: Causally disconnected regions of the universe appear similar because they decompress from the same informational seed.

Accounts for CMB uniformity: The cosmic microwave background's remarkable uniformity reflects the underlying informational architecture from which everything decompressed.

Predicts structure formation: Large-scale structures weren't generated randomly from quantum fluctuations—their patterns were encoded in the compressed information and unfold during decompression.

4.2 Information as Cause, Mass as Effect

This framework fundamentally inverts the causal chain:

Standard cosmology:

Initial mass-energy → expands → cools → forms structures → generates complexity

Information decompression cosmology:

Initial information (compressed) → decompresses → manifests as spacetime → generates mass-energy through information instantiation → mass-energy organizes according to informational constraints (gravity) → complexity emerges as encoded patterns instantiate
Mass-energy is not the starting point—it's the product of information becoming manifest in physical form. This is why mass "resists" expansion while information drives it. Mass is downstream from information, not upstream.

Information as the weaving force: The needle (information's organizational logic) guides the thread (information's specific patterns) through energy-space, creating the intricate fabric of physical reality. The fabric exists because information weaves it into being.

4.3 Dark Energy as Decompression Momentum

Current cosmology struggles to explain dark energy—the mysterious force causing accelerating expansion. In the information decompression framework:

Dark energy is the ongoing decompression dynamic itself.

The universe continues to expand because the information is still unpacking. Acceleration occurs because decompression follows an algorithmic trajectory encoded at inception. We don't need to invoke a mysterious repulsive force—we need only recognize that information decompression has momentum.

Mathematically, if expansion rate $H(t)$ correlates with information instantiation rate dI_{inst}/dt :
 $H(t) \propto dI_{\text{inst}}/dt$

The acceleration we observe reflects that information instantiation is itself an accelerating process: more complex structures enable even more complex emergent phenomena, which require faster decompression rates to instantiate.

Recent observational support: 2025 studies, including analyses from Yonsei University and JWST data, suggest the universe's expansion may be decelerating rather than accelerating, challenging traditional dark energy models. This aligns with our framework: if decompression is a finite process, momentum could wane as unpacking nears completion, leading to slowdown. Future observations could test this by correlating deceleration onset with complexity plateaus.

5. Dual Dynamics: Global Decompression, Local Optimization

5.1 The Beautiful Symmetry

Vopson's gravity (local information entropy reduction) and cosmic expansion (global information decompression) create complementary dynamics:

Global scale:

Information decompresses → entropy increases → expansion

continues Total information entropy of the universe increases over time

Local scale:

Gravity clusters matter → information entropy decreases → structures form

Local information entropy in gravitationally bound systems decreases over time

This isn't a contradiction—it's a dual dynamic:

The universe globally "spreads out" its information (expansion)

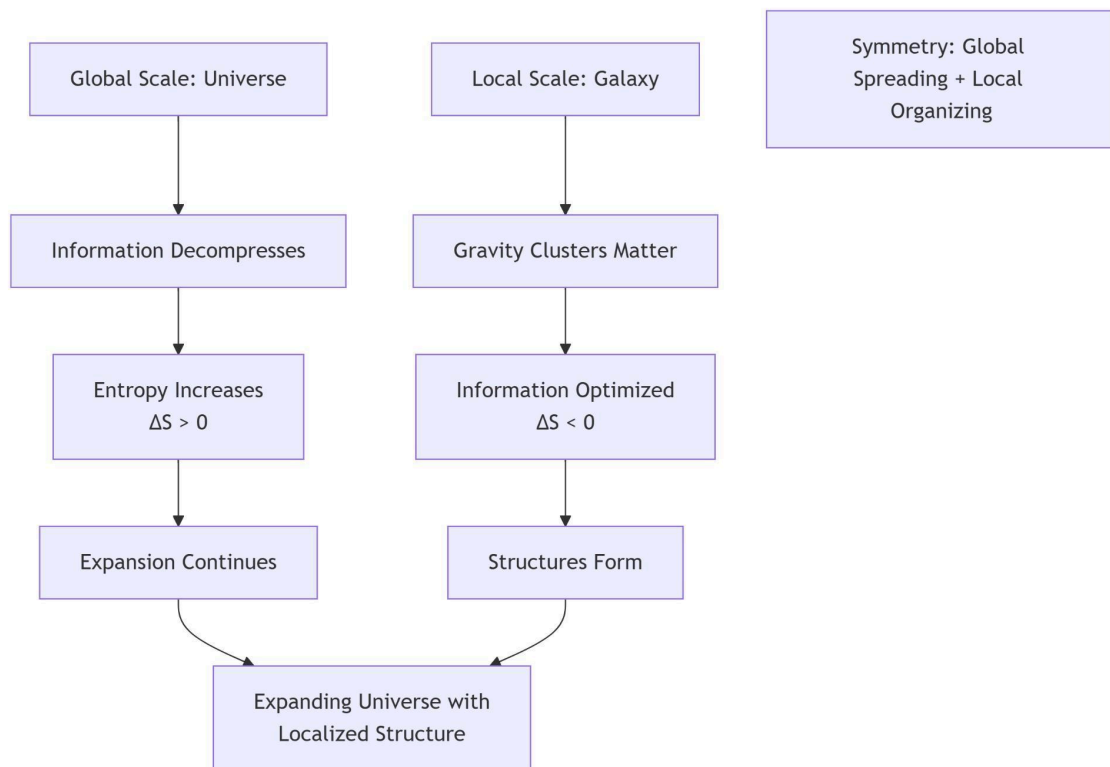
While locally "organizing" information efficiently (gravitational structure formation)

5.2 Structure Formation as Information Clustering

Galaxy formation, star formation, and planetary systems emerge from gravity's information optimization:

Initial density fluctuations (encoded in the decompression algorithm)

Gravitational attraction clusters matter (reduces local information entropy)



Structures form at multiple scales (fractal organization reflects optimal information processing)

The cosmic web emerges (network topology minimizes information complexity)

The reason the cosmic web resembles neural networks and mycelial networks (Vazza & Feletti, 2020; Carboni, 2025) becomes clear: all are manifestations of the same informational optimization principles.

Network architecture emerges repeatedly because it represents the most efficient structure for information processing under physical constraints. This isn't coincidence or metaphor—it's the same informational grammar operating at different scales.

5.3 Mathematical Representation

For a gravitationally bound system (galaxy, star cluster):

$\Delta S_{\text{local}} < 0$ (local entropy decreases)

$\Delta I_{\text{local}} < 0$ (local information entropy decreases, per Vopson)

For the expanding universe globally:

$\Delta S_{\text{global}} > 0$ (global entropy increases)

$\Delta I_{\text{global}} > 0$ (global information entropy increases through decompression) The universe simultaneously compresses information locally (structure) while decompressing information globally (expansion). This dual process drives cosmic evolution from the Big Bang through structure formation to biological complexity.

6. From Cosmic Structure to Consciousness

6.1 Chemistry as Information Instantiation

When gravitational clustering produces stars, nuclear fusion creates heavy elements. Each element represents instantiated information about atomic structure. Chemistry becomes possible when information decompression reaches atomic complexity.

The periodic table is not arbitrary—it reflects informational constraints encoded at cosmic inception. Atomic structure, bonding rules, and molecular possibilities were implicit in the compressed information. When stars form and forge heavy elements, they're not creating something new—they're instantiating patterns that were already present as informational potential.

6.2 Life as Self-Replicating Information Processing Life emerges when:

Chemical complexity crosses a threshold of organizational sophistication

Autocatalytic cycles form (information processing loops)

Replication mechanisms instantiate (information copying with variation)

Evolution begins (exploration of morphospace encoded in initial information)

As I argued in my previous work (Carboni, 2025), biological evolution shows directional complexity increase not because natural selection "drives toward" complexity, but because evolution explores a structured possibility space—morphospace—that was encoded in the initial informational compression.

Evolution doesn't create complexity from nothing. It discovers complexity patterns already present in the informational architecture, navigating toward attractor states that represent stable, efficient organizational forms.

6.3 Consciousness as Self-Examining Decompression

Consciousness emerges when information integration reaches a specific threshold—what Integrated Information Theory (Tononi et al., 2016) quantifies as Φ (integrated information).

At this threshold:

The system processes integrated information about its own information processing (recursive self-reference)

Subjective experience is what high-integration information dynamics "is" from the inside

Consciousness represents the decompressed information recognizing its own patterns.

Within this framework, the hard problem of consciousness is reframed rather than demonstrably dissolved. There is no gap between physical processes and subjective experience because there aren't two different things. There is only information at different scales of integration.

Third-person description (neurons, networks, information flow) and first-person description (experience, qualia, what-it's-like) are two observational frames on the same informational dynamics—like describing water as H₂O molecules (objective frame) versus experiencing wetness (subjective frame). Both are equally valid perspectives on one phenomenon.

Humans represent a point where the cosmic decompression has achieved sufficient local integration to examine the decompression process itself. We are the universe reading its own code.

6.4 The Constraint Gradient: Why We Can Simulate What We Cannot Build

Here's a striking asymmetry that illuminates the architecture of information: we can simulate fire-breathing dragons in digital worlds, yet we cannot build them in flesh. We can program worlds with impossible physics, yet cannot construct them materially. Why?

Information becomes progressively more constrained as it instantiates through successive layers of the decompression hierarchy.

At the foundational level, information has the widest range of expression permitted by the ontology of the cosmos, but it is not unconstrained. Its unfolding is governed from the outset by a grammar: the structure of reality that permits some relations, transformations, and trajectories while excluding others. As information instantiates through quantum,

physical, chemical, and biological regimes, additional constraints accumulate and further narrow the available possibility space. Consider the hierarchy:

SINGULARITY → Widest grammatically permitted informational potential

MATHEMATICS → Abstract formal expression under logical and structural constraint

QUANTUM MECHANICS → High flexibility (probabilistic, observer-relative)

PHYSICAL LAWS → Moderate constraint (constants fixed)

CHEMISTRY → Heavy constraint (periodic table, bonding rules)

BIOLOGY → Very heavy constraint (chemistry + evolution)

COMPLEX LIFE → Extreme constraint (all previous layers)

Each layer inherits and adds to the constraints of previous layers. The deeper into instantiation, the more restricted the possibility space.

The dragon example:

Possible in mathematics: We can model dragon flight, fire thermodynamics (equations work)

Possible digitally: We can render it in simulations (video games do this constantly)

Impossible chemically: Biological tissue cannot coexist with sustained combustion at dragon-fire temperatures

Impossible biologically: Evolution cannot reach that configuration given thermodynamic constraints

Why simulations "work" at higher layers:

Digital simulation operates through a physical computational substrate while representing rule systems that need not reproduce every constraint governing the represented object.

When we program a virtual world, we can specify alternative internal rules because symbolic representation is not identical to material construction. The computer itself remains fully constrained by the physics of this universe; what is loosened is the represented model, not reality's grammar.

A simulated dragon need not be assembled from living tissue capable of surviving combustion. The simulation therefore avoids the construction constraints of the represented organism, even though the hardware executing the simulation remains chemically and physically constrained. Informational patterns are manipulated through an instantiated substrate; they do not float free of ontology.

Mathematics as privileged access:

This explains the "unreasonable effectiveness of mathematics" in physics—a puzzle that has troubled philosophers of science since Wigner (1960). One possibility within this framework is that mathematics is effective because it articulates highly abstract relations and invariances that remain visible across many instantiated domains. This does not make mathematics unconstrained or place mathematical thought outside physical reality; it makes mathematics a powerful language for describing recurrent structural relations within the grammar of reality.

Physical laws are early-stage instantiations of informational patterns, still close enough to the root that mathematical elegance and universality remain. Pure mathematics is "cleaner" than chemistry precisely because it hasn't been "made messy" with the accumulated constraints that chemistry inherits from physics.

Consciousness accesses multiple layers:

Human consciousness demonstrates a remarkable feature: we can imagine patterns that are chemically or physically impossible to construct. We can conceive of fire-breathing dragons, four-dimensional objects, infinite sets, alternative physics.

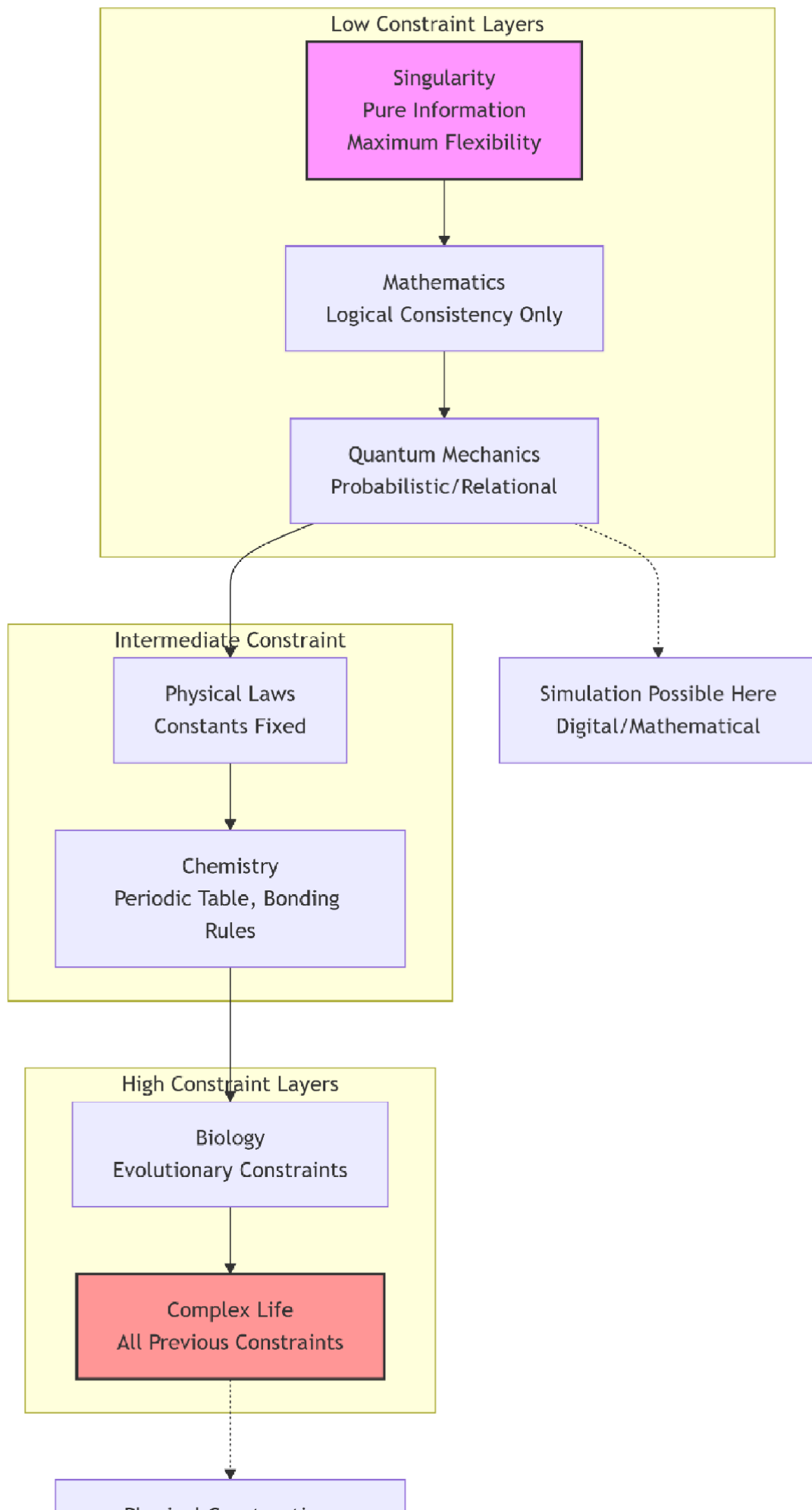
This suggests that consciousness can represent possibilities that cannot be physically constructed by the organism that represents them. That asymmetry does not require

consciousness to escape its biological substrate or literally access a higher ontological layer. A constrained physical system can encode, recombine, and model descriptions of states that are not themselves physically realizable. We can think beyond what we can build because representation and construction are differently constrained activities. Neural processes are constrained by chemistry—neurons are biological machines operating under thermodynamic limits. But consciousness as integrated information can represent patterns from less-constrained layers of the informational hierarchy.

This is why mathematics, art, philosophy, and imagination feel "purer" or more "free" than physical construction. They operate where information retains greater flexibility, before the full weight of accumulated constraints bears down.

The cosmos trades freedom for existence. Fundamental information may possess a wider range of permitted expression than deeply instantiated physical forms, but it is never arbitrary or grammarless. Physical reality actualizes specific possibilities within the constraint architecture of the cosmos. Yet consciousness, emerging at the edge of this process, retains access upward—toward the freer informational layers—even while instantiated in constrained flesh.

We are materially instantiated beings capable of representing possibilities beyond those we can construct.



6.5 Dreams: Simulation as Native Function

Every night, human consciousness spontaneously generates convincing simulated realities—complete with autonomous-seeming entities, physical environments, and narrative coherence—without external sensory input and without technological apparatus.

Dreams reveal that consciousness operates simultaneously at multiple informational layers. During dreams, consciousness generates experience internally, without requiring the dreamed objects to be materially constructed in the external environment. The dreaming process itself remains embodied in neural, chemical, and physical activity; what differs is the mode of representation, not exemption from reality's constraints.

You dream a dragon and experience it immediately—no need to build one from atoms.

What dreams demonstrate:

Generative capacity: Dreams create novel scenarios beyond memory replay—impossible physics, non-existent people, contradictory situations. This shows that consciousness can generate flexible internal representations under conditions of reduced external sensory constraint.

Phenomenological realism: Dream experiences feel real during the dream state, achieving subjective authenticity indistinguishable from waking experience—until lucidity or waking reveals the simulation.

Autonomous entities: Dream characters behave as if independently conscious—they surprise us, argue with us, express emotions. They are convincing simulations of other minds generated entirely within our integrated information processing.

Lucidity phenomena: When dreamers achieve lucidity (recognizing "this is a dream"), dream characters sometimes react to this revelation—expressing sadness, pleading not to be dismissed, or simply vanishing. This suggests simulated entities achieve sufficient informational coherence to exhibit quasi-autonomous responses.

Phylogenetic distribution:

All mammals exhibit REM sleep and behavioral evidence of dreaming. Birds show similar patterns. This suggests dreaming—and therefore simulation capacity—is ancient and widespread.

Simulation sophistication scales with integrated information (Φ):

Low Φ : No dreaming (insects, simple organisms)

Medium Φ : REM sleep, possible memory consolidation (most mammals)

High Φ : Generative dreaming, novel scenarios, lucidity potential (humans, possibly great apes, corvids, cetaceans)

To be conscious is to simulate. Higher integration enables more sophisticated internal modeling.

Across cultures, dreams are documented to facilitate problem-solving (e.g., Kekulé's benzene structure, Mendeleyev's periodic table) through pattern recognition. This suggests consciousness during sleep—operating at less-constrained informational layers—can detect subtle signals missed in waking states.

What this reveals about information:

The widespread, spontaneous nature of dreaming supports the more modest claim that internally generated world-modeling is a native capacity of many conscious organisms rather than a technological invention. Within this framework, that capacity is interpreted as one expression of informational organization, not as proof that consciousness escapes physical constraint.

If we spontaneously generate simulated realities within our own integrated information processing, this suggests reality and simulation share a substrate. The difference is hierarchical (different instantiation layers), not ontological (different kinds of stuff). We are dreamers who might themselves be dreamed. Both the dreamer and the dream are real informational patterns.

6.6 Cross-Substrate Information Transfer: Empirical Evidence

One of the most compelling pieces of evidence for information as a substrate-independent principle comes from observing information transfer and coordination across radically different physical substrates.

The Chick-Robot Experiment:

In experiments where newly hatched chicks imprint on small mobile robots, researchers observe remarkable phenomena:

Chicks (biological substrate: neurons, neurotransmitters, evolved wetware) form attachment bonds with robots (mechanical/computational substrate: motors, sensors, simple algorithms)

The robots' behavior patterns change based on chick proximity and behavior

Feedback loops emerge that produce coordinated behavior between biological and artificial systems

The chick-robot pair achieves a form of mutual information exchange and behavioral synchronization

The profound implication: A biological neural network and a simple computational system—radically different physical implementations—can exchange information effectively, establish meaningful feedback loops, generate coordinated patterns, and form what resembles "relationship" or mutual adaptation.

If information were substrate-specific—if biological information and computational information were fundamentally different kinds of things—this cross-substrate coordination would be mysterious. But if information is substrate-independent at a fundamental level (as our framework proposes), then biological and computational systems can interface naturally because they're both instantiating informational patterns.

The chick's neurons and the robot's circuits are different "fabrics," but the informational "needle" can weave patterns that span both.

DishBrain: Direct Empirical Evidence:

One of the most striking empirical demonstrations of information's fundamental nature comes from neurons in a dish learning to play Pong (Kagan et al., 2022).

Researchers cultured approximately 800,000 neurons on a multielectrode array, provided electrical feedback about paddle position and ball location, and observed that within five minutes, the neurons learned to keep the ball in play—demonstrating goal-directed, adaptive behavior.

These neurons had no organized brain structure, no body or sensorimotor coordination, no evolutionary history of game-playing, no prior training examples, and no complex architecture beyond basic connectivity.

Yet they discovered how to minimize prediction error and achieve a goal that had never existed in their evolutionary history.

The profound question: How did they learn? Where did the "know-how" come from?

Traditional answer: "Reinforcement learning—feedback shaped synaptic weights through plasticity mechanisms."

But this describes the mechanism, not the source. Why does synaptic plasticity follow patterns that lead to goal achievement? Why does minimizing prediction error constitute

learning? Why do these neurons, with no evolutionary preparation for this task, spontaneously self-organize toward optimization?

Information framework answer:

The neurons exhibited information-processing capacities made possible by their material organization and by the lawful structure of the reality in which that organization exists. Within this framework, learning is interpreted as a constrained reorganization of informational relations under feedback. The experiment does not by itself prove a primordial informational substrate, but it is consistent with the broader claim that informational organization can be causally efficacious across differently arranged systems.

Consider the parallel with Vopson's gravity: gravitational attraction minimizes information entropy by clustering matter. Neural learning minimizes prediction error by adjusting synaptic weights.

These are the same optimization principle—information self-organizing toward efficient representation—instantiated at different scales.

What this supports, more cautiously:

If 800,000 neurons with no biological scaffolding can learn a novel task through informational feedback alone, this demonstrates:

Information processing capacity is intrinsic to substrate, not emergent from complexity:

Simple neuronal networks exhibit goal-directed learning without requiring organized brain structure.

Goal-directed behavior emerges spontaneously when feedback is provided: The neurons weren't programmed to play Pong. They discovered how through informational feedback about system state.

No special evolutionary history required: These neurons had no ancestors that played games, yet they learned. Learning capacity is fundamental to information-processing substrates.

Learning is pattern discovery, not pattern creation: The neurons discovered optimization strategies available as informational patterns in possibility space.

Additional Cross-Substrate Examples:

The pattern extends across multiple domains:

Brain-computer interfaces: Neural signals (biological) to digital commands (computational).

Paralyzed patients control robotic limbs through thought alone.

DNA to RNA to Protein: Information traversing radically different chemical substrates while maintaining coherent meaning.

Quantum computing reading classical data: Information maintaining coherence while moving between quantum and classical regimes.

Artificial pollination of orchids: Mechanical devices successfully trigger plant reproductive responses by providing appropriate informational cues (mechanical pressure patterns, chemical signals).

All these examples show information maintaining coherence and causal efficacy while moving between radically different physical substrates.

Theoretical Implications:

If information can drive coordinated behavior across biological-mechanical boundaries (chick-robot), chemical-digital boundaries (DishBrain), quantum-classical boundaries (quantum computing interfaces), and neural-computational boundaries (brain-computer interfaces), this suggests information really is a substrate-independent organizing principle, not an epiphenomenon of particular physical arrangements.

This directly supports our cosmological claim: If information can organize and coordinate across such diverse substrates today, it's plausible that information could be the primary organizing principle at cosmic inception—the "needle" that begins weaving energy into structured spacetime.

The chick-robot and DishBrain experiments aren't just interesting neuroscience—they're empirical evidence that information operates as a fundamental organizing force that transcends any particular physical implementation.

The DishBrain experiment demonstrates that informational feedback can participate causally in the reorganization of material systems. Neurons physically reorganized their synaptic connections in response to informational feedback. Information wasn't epiphenomenal—it was the driving force directing material change.

On the ontology proposed here, brains are not merely containers that happen to process an otherwise secondary abstraction called information. They are biological instantiations of informational dynamics operating under physical, chemical, and biological constraints.

Different substrates may support different forms of information processing, but no instantiated process is thereby exempt from the grammar of reality.

6.7 Distinguishing From Simulation Hypothesis

This framework is sometimes confused with the simulation hypothesis (Bostrom, 2003)—the proposition that our universe is a computer simulation run by advanced beings in some "base reality." The positions are fundamentally different.

Simulation hypothesis problem:

The question "Are we in a simulation created by beings in base reality?" faces insurmountable conceptual problems:

What is being simulated? If our universe simulates physics, where do those physical laws come from?

Are the simulators also simulated? This leads to infinite regress or requires arbitrary "base reality."

Where do the laws being simulated originate? They exist as informational patterns regardless of instantiation.

Our resolution:

We CAN create simulations that feel real not because we're copying some "more real" reality, but because both physical reality and digital simulation instantiate the same informational patterns—they're built from the same substrate.

Consider:

Physical universe: Information decompressing from maximal compression, instantiating as spacetime, matter, energy

Digital simulation: Information instantiating through computational processes

Dreams: Information instantiating through neural processes

Each is a real informational process in a different sense and at a different level of instantiation, but they should not be collapsed into equivalence. A physical cosmos, a digital model, and a dream differ in causal independence, substrate, persistence, and constraint.

Their comparability lies in informational organization, not in the claim that they are ontologically interchangeable.

What makes our universe NOT "a simulation":

Our framework does not require the physical universe to be a simulation in the usual sense: Not copying anything: The initial informational compression didn't replicate some other universe. It IS the universe—the primary instantiation. There's nothing, it's a simulation of "of."

Not externally created: No programmer wrote the laws of physics as one writes code. The laws emerge from informational constraints encoded at cosmic inception.

Not less real: There's no "more real" reality that our universe is a pale copy of. This is the instantiation of information itself.

Why simulations work:

When we simulate physics in computers, we instantiate the informational patterns of physical law. A physics simulation doesn't create "fake gravity"—it instantiates gravity's informational pattern ($F = Gm_1m_2/r^2$ and relativistic extensions) in computational substrate. That's why it accurately predicts outcomes: it's running the same algorithm on different hardware. Success proves information is substrate-independent, not that we're "probably simulated."

The hierarchical relationship:

Information Substrate → Cosmic Decompression → Quantum Mechanics → Classical Physics → Chemistry, Biology → Consciousness → Technology → Digital Simulations → Nested Simulations

Each level instantiates informational patterns. None is "fake"—they're genuinely real patterns at different hierarchical positions.

If future civilizations create conscious simulations:

Those simulated beings would be:

Genuinely conscious (if achieving sufficient Φ)

Really experiencing qualia (integration is real regardless of substrate)

Actually existing as patterns (they're genuine informational entities)

They wouldn't be "fake people in a fake world." They'd be people whose informational instantiation happens through computation rather than cosmic decompression.

The logical error of the simulation hypothesis becomes clear through analogy: If I copied a Rembrandt painting so perfectly that it was indistinguishable from the original, would that make the original Rembrandt a copy too? Obviously not. Yet this is precisely the logic simulation hypothesis employed.

We've developed the ability to simulate physics computationally—to instantiate the same informational patterns ($F=ma$, $E=mc^2$, Maxwell's equations) in silicon that the universe instantiates in spacetime. Because we can copy reality's patterns, the simulation hypothesis concludes we ourselves might be copies. But the ability to copy doesn't make the original a copy. It demonstrates that informational patterns are substrate-independent—they can be instantiated multiply without any instantiation being "less real" than others.

The question "What does the simulation hypothesis simulate?" exposes the confusion. If you answer "it simulates physics," then you're admitting physics—the informational patterns—exists independently as something to be simulated. The simulation is copying patterns that exist in the informational substrate. Our universe doesn't simulate those patterns; it IS those patterns instantiating for the first time in cosmic decompression. Creating a simulation proves information is substrate-independent. It doesn't prove we're in someone else's simulation any more than painting a perfect copy of the Mona Lisa proves the original Mona Lisa is itself a copy.

The profound insight:

The takeaway isn't "we might be simulated" but: "Everything is information, instantiable in multiple ways, and consciousness arises when information integration crosses thresholds regardless of substrate."

We are not in a simulation. We are not in a base reality. We are in information, examining information, made of information—patterns recognizing patterns.

7. Empirical Predictions

Empirical Predictions of Decompression Framework



This framework generates testable predictions distinct from standard cosmology.

7.1 Cosmological Predictions

Prediction 1: Structured CMB Patterns

If expansion reflects information decompression from an encoded initial state, the cosmic microwave background should contain structured information patterns beyond random quantum fluctuations.

Specific prediction: Higher-order correlations in CMB temperature fluctuations should reveal algorithmic structure rather than pure randomness.

Test: Advanced statistical analysis of Planck satellite data or future CMB missions, looking for non-random information-theoretic patterns in temperature fluctuations. Apply algorithmic information theory metrics (Kolmogorov complexity, mutual information measures) to CMB data. A purely random, Gaussian CMB field would support standard inflation from quantum fluctuations. The discovery of non-random, algorithmic patterns (e.g., low Kolmogorov

complexity in anisotropy maps) would be direct evidence for a decompression from an informationally structured initial state.

Prediction 2: Expansion-Information Correlation

Expansion acceleration should correlate with measures of information instantiation in the universe (structure formation rate, complexity increase).

Test: Correlate cosmic expansion rate evolution with the universe's integrated information content over time, measured through structure complexity metrics. As the universe forms more complex structures (galaxies, galaxy clusters, large-scale filaments), expansion dynamics should show corresponding changes.

Prediction 3: Gravitational Information Optimization

Per Vopson, gravitationally bound systems should show measurable information entropy reduction.

Specific prediction: Information content of a galaxy cluster should be lower than the sum of information content of its constituent galaxies measured independently.

Test: Develop information entropy metrics for astronomical structures and measure before/after gravitational binding. Compare information content of diffuse gas clouds versus formed stars/galaxies.

Prediction 4: Expansion Deceleration and Complexity Plateaus

If decompression is finite, expansion should eventually slow as unpacking nears completion. This slowdown should correlate with plateaus in cosmic complexity increase.

Test: Monitor expansion rate evolution alongside structure formation rates. Recent 2025 studies suggesting expansion slowdown could be early evidence. Look for correlation between deceleration timing and reduced rates of new structure formation.

7.2 Quantum and Fundamental Physics

Prediction 5: QM as Informational Grammar

If quantum mechanics describes informational relationships, quantum behaviors should follow information-theoretic principles more closely than classical mechanical principles.

Test: Quantum outcomes should be better predicted by information entropy measures, mutual information, and complexity metrics than by geometric or mechanical models.

Entanglement should scale with shared information history rather than spatial proximity.

Prediction 6: Quantum Gravity Information Substrate

If information is fundamental, quantum gravity should reveal informational rather than purely geometric structure at Planck scale.

Test: Look for discrete information units (bits, qubits) in quantum gravitational experiments. Loop quantum gravity and causal set theory predictions should show information-theoretic structure. Gravitational wave signatures might reveal discreteness consistent with informational "pixels."

Prediction 7: Consciousness Quantum Coherence

Neural systems approaching consciousness thresholds should exhibit quantum coherence patterns reflecting integrated information processing.

Test: Measure quantum coherence times and entanglement patterns in neural tissue across organisms of varying consciousness levels. High- Φ systems should maintain quantum information integration longer than classical decoherence models predict.

7.3 Biological and Consciousness Predictions

Prediction 8: Morphospace Boundaries

Biological forms explore a bounded possibility space encoded in initial information. Novel synthetic organisms should converge on attractor states rather than exploring randomly.

Test: Xenobots, organoids, and synthetic biology experiments should show constrained

form-space exploration with predictable convergence patterns. Track morphological diversity in artificial life systems—should see boundaries emerge despite high degrees of freedom.

Prediction 9: Consciousness Integration Thresholds

Consciousness should correlate with specific integrated information (Φ) thresholds, showing discontinuous emergence rather than smooth scaling with neural complexity.

Test: Measure Φ across organisms and artificial systems; consciousness indicators should appear above specific threshold values. Simple neural networks below threshold should show no subjective markers; systems above threshold should exhibit qualitatively different behaviors (attention, learning flexibility, novel problem-solving).

Prediction 10: DishBrain Scaling

Simple neuronal networks should exhibit learning capacity proportional to connectivity and feedback quality, independent of biological context or evolutionary history.

Test: Vary neuron count, connection density, and feedback mechanisms systematically. Learning should scale predictably with information-processing capacity metrics. Test with neurons from different species/tissue types—learning capacity should depend on network parameters, not biological origin.

Prediction 11: Cross-Substrate Information Transfer

Information coordination should be achievable across any sufficiently connected substrates that can exchange signals, regardless of physical implementation differences.

Test: Extend chick-robot and DishBrain paradigms to new substrate combinations (plant-mechanical, quantum-classical hybrid systems, chemical-computational interfaces).

Successful coordination across arbitrary substrate pairs would support substrate-independence of information.

7.4 Constraint Gradient Verification

Prediction 12: Systematic Constraint Increase

Degrees of freedom should decrease monotonically with instantiation depth across the hierarchy from mathematics through biology.

Test: Develop information-theoretic metrics measuring possibility space at each layer. Should show systematic reduction: mathematics > quantum mechanics > physics > chemistry > biology. Measure combinatorial explosion of possibilities at each level—should see progressive restriction.

Prediction 13: Simulation-Construction Gap

The gap between what can be simulated versus physically constructed should systematically increase deeper in the constraint hierarchy.

Test: Catalog what's simulatable at various computational levels versus what's constructible with current/theoretical technology. The gap should be minimal for abstract mathematics, maximal for complex biology. As simulation technology improves, the gap should narrow for higher layers but remain large for deeply constrained layers.

7.5 What Would Falsify This Framework

Critical falsifiers:

Mass-only expansion: Demonstration that expansion is definitively caused by mass-energy dynamics with no informational component.

Non-informational gravity: Evidence that gravity has no informational character, contradicting Vopson's findings.

Random CMB: Cosmic microwave background showing purely random patterns with no encoded structure beyond quantum noise.

Low- Φ consciousness: Consciousness consistently emerging in systems with low information integration or high decomposability.

Random morphospace: Biological evolution showing completely random morphospace exploration with no attractor dynamics or constraint structures.

Substrate failure: DishBrain-type experiments failing to replicate learning across different substrates with similar connectivity.

No constraint gradient: If degrees of freedom don't decrease with instantiation depth, or if simulation-construction gap doesn't correlate with constraint layers.

These are genuine empirical tests. The framework stands or falls on observational evidence, not philosophical preference.

8. Advantages Over Standard Cosmology

8.1 Explanatory Parsimony

Standard cosmology requires:

Inflaton field (never directly observed, properties fine-tuned)

Dark matter (never detected despite decades of searching)

Dark energy (mysterious repulsive force with no mechanism)

Fine-tuned initial conditions (requiring anthropic explanation)

Separate explanations for gravity, expansion, structure, life, consciousness

Information decompression framework:

Single substrate (information) with minimal mass

Gravity emerges from information optimization (Vopson, empirically supported)

Expansion is information decompression momentum (natural mechanism)

Structure formation is local information clustering (follows from gravity)

Life and consciousness are progressive information integration (continuous with physics)

Initial conditions encode the algorithm (fine-tuning explained by informational structure)

One explanatory principle instead of five separate mechanisms. That's scientific parsimony.

8.2 Resolves Major Paradoxes

Horizon Problem: Standard cosmology struggles to explain why causally disconnected regions are so similar. Our framework: They decompress from the same informational seed, so similarity is expected.

Flatness Problem: Why is spatial curvature so close to flat? Our framework: Flatness reflects optimal information decompression geometry encoded in the initial structure.

Monopole Problem: Predicted magnetic monopoles don't appear. Our framework: They're not encoded in the decompression algorithm, so they don't instantiate.

Arrow of Time: Why does time have a direction? Our framework: Time is the dimension of information unpacking—past = compressed, future = decompressed. Directionality is built in.

Infinite Density Singularity: Standard cosmology breaks at infinite density. Our framework: Maximal information density with minimal mass—no infinite mass required.

Fine-Tuning Problem: Why are physical constants precisely calibrated for complexity? Our framework: They're parameters of the decompression algorithm, inherently structured for complexity emergence.

8.3 Unifies Domains

This isn't just cosmology—it's a unified framework spanning:

Physics: Gravity (optimization), expansion (decompression), quantum mechanics (grammar), spacetime (instantiation)

Biology: Evolution (morphospace exploration), complexity increase (attractor dynamics), network structures (optimization principles)

Consciousness: Emergence (integration thresholds), qualia (first-person information dynamics), learning (pattern discovery)

Technology: Simulation (alternative instantiation), AI (information processing), cross-substrate interfaces (substrate independence)
Standard cosmology offers no connection between cosmic expansion and consciousness. Our framework shows they're different scales of the same informational process. Quantum mechanics stops being a mystery requiring interpretation—it's the grammar of information itself. The "weird" behaviors (superposition, entanglement, uncertainty) are features, not bugs. They're the logic of informational relationships within a decompressing network.

This points toward quantum gravity reconciliation: If both general relativity and quantum mechanics emerge from informational dynamics, they share a common foundation. Gravity optimizes local information (Vopson), quantum mechanics governs how information relates and updates, and spacetime manifests as the relational structure of information.

Resolution of dark matter and dark energy: These could represent informational effects—either uninstantiated information influencing spacetime geometry, or diffuse information not yet organized into recognizable particle forms. Rather than searching for exotic particles or mysterious forces, we might be observing the informational substrate itself.

We don't live in a universe that has information. We live in a universe that is information.

9. Philosophical Implications

9.1 Information as Ontological Primitive

This framework is monistic—information is fundamental. But it avoids reductionism. Consciousness, life, and meaning aren't illusions or "nothing but" simpler processes. They're genuine features of informational architecture with their own causal efficacy.

Higher-level patterns (organisms, ecosystems, consciousness) are real even though they supervene on simpler information dynamics. The pattern is ontologically real, not merely a convenient description.

Quantum mechanics supports this ontology. The observer-relative informational interpretation (Rovelli, Zeilinger, Brukner) shows that quantum "stuff" isn't fundamental—informational relationships are. Particles, fields, and spacetime are patterns in a relational informational network, not autonomous entities with intrinsic properties.

9.2 Observer-Relative Quantum-Classical Divide Dissolves

The measurement problem and quantum-to-classical transition stop being mysteries.

There's no fundamental divide between quantum and classical realms—only different scales of information integration:

Quantum regime: Low-integration systems where informational relationships remain observer-relative and context-dependent.

Classical regime: High-integration systems where information has been sufficiently integrated that observer-relative differences become negligible.

Consciousness sits at a unique threshold: high enough integration to experience unified qualia, yet quantum enough to maintain informational flexibility. This might explain why consciousness seems to straddle quantum and classical descriptions—it operates at the transition scale.

9.3 Time as Information Decompression

If the universe is fundamentally information decompression, time gains a new interpretation.

Within this speculative framework, time may be interpreted as the ordered dimension of informational unfolding:

Past = more compressed informational state

Present = current state of partial decompression

Future = continued unpacking toward fuller instantiation

This explains why time appears irreversible and why we experience temporal flow—we're embedded in an asymmetric decompression process. Entropy increase isn't imposed by thermodynamic law alone; it reflects the directionality of decompression.

A stronger future version of the theory would need to determine whether quantum indeterminacy can legitimately be modeled as incompletely instantiated informational structure. The present framework should not simply identify measurement with 'premature instantiation': that claim exceeds what has been derived and must remain a hypothesis rather than a conclusion.

9.4 The Boundary of Inquiry

We cannot answer: "Why this particular informational compression?"

That question may be empirically unanswerable. We're inside the decompressing system, made of its patterns, with no external vantage point. This is the epistemic limit of the framework.

But we can answer: "Given this informational architecture, what follows?"

The trajectory from singularity to consciousness follows with compelling logical necessity. That's the domain where we can make progress.

Quantum mechanics reveals an additional boundary: we cannot access "absolute" information, only observer-relative information relative to our observational perspective within the network. This isn't a limitation of our instruments—it's a fundamental feature of informational ontology. Information is inherently perspectival.

9.5 Meaning as Discovery, Not Construction

If patterns pre-exist in the compressed information, several implications follow:

Mathematics describes reality because mathematical structures are aspects of the informational substrate, not human inventions imposed on nature.

Scientific theories are discoveries of encoded patterns, which is why they feel like uncovering something that was always there.

Art renders legible the fundamental structures of existence—compression, emergence, pattern, dissolution—which is why profound art feels like revelation rather than invention.

Consciousness is meaning recognizing itself—the universe achieving sufficient integration to examine its own patterns.

This provides grounding for meaning without teleology. The universe isn't going toward an external purpose, but it has intrinsic structure that matters. Patterns are real, and recognizing patterns is what consciousness does.

10. Open Questions and Future Research

Can we calculate I_{max} ? What was the initial information content at the singularity?

Determinism or probability? Is decompression deterministic or probabilistic?

What is the endpoint? Will decompression complete, or continue indefinitely?

Quantifying the constraint gradient: Can we develop precise information-theoretic metrics?

Computational modeling: Can we build abstract information decompression models that produce cosmic structure patterns?

Multiverse implications: If multiple informational compressions are possible, does this imply a multiverse?

Artificial consciousness: Can silicon-based systems achieve genuine consciousness?

Cambrian explosion: Does the constraint gradient explain rapid morphological diversification?

Simulation-construction gap: Can we systematically catalog what's simulatable versus constructible?

Cross-scale optimization: Are there universal optimization principles across all scales?

Information mass equivalence: Can we establish precise relationships between information content and physical mass/energy?

Substrate transfer limits: Are there fundamental limits to cross-substrate information transfer?

11. Conclusion

We've argued that information—as a primordial organizing principle or logical structure—drove cosmic expansion. The initial singularity contained maximal compressed information density with minimal mass. The Big Bang was information decompression—the execution of this logic—unleashing structured expansion.

This framework resolves major cosmological paradoxes, unifies physics with biology and consciousness, generates testable empirical predictions, and provides philosophical grounding for meaning without teleology.

Information's inherent lightness—the absence of inertial resistance in a pre-physical rule set—makes it the ideal driver for explosive expansion. Mass-heavy structures are what information produces as it decompresses, not what causes expansion. Information is the cause; mass is the effect.

The needle-and-fabric metaphor captures the relationship: information (needle/logic) guides energy (fabric) into structured forms. The weaving requires work (Landauer costs), but the pattern itself is lightweight and can move rapidly, enabling explosive cosmic expansion. We are local regions where the decompression has achieved sufficient integration to examine itself. In consciousness, the universe reads its own code.

This framework is necessarily speculative, proposing an ontological inversion of standard physical models. Its value lies in its explanatory parsimony, its unification of disparate domains, and its generation of novel, testable predictions. The primary challenge ahead is the formal mathematical derivation of spacetime geometry from information-theoretic first principles—a challenge akin to deriving thermodynamics from statistical mechanics.

The decisive qualification is that decompression is not arbitrary release. Information is fundamental and creatively active, but its unfolding is always constrained by the grammar of reality. The universe therefore does not move from unlimited informational freedom to later physical restriction. It moves from a widest grammatically permitted potential through successive regimes of further specification and accumulated constraint. This distinction between information as active dynamic and grammar as the structure of permitted unfolding is essential to the ontology defended here.

References

Bostrom, N. (2003). Are you living in a computer simulation? *The Philosophical Quarterly*, 53(211), 243–255.
<https://doi.org/10.1111/1467-9213.00309>

Brukner, C. (2017). On the quantum measurement problem. In R. Bertlmann & A. Zeilinger (Eds.), *Quantum [Un]Speakables II* (pp. 95–117). Springer.
https://doi.org/10.1007/978-3-319-38987-5_5

Carboni, D. (2025). Informational inevitability: The deep chained nexus from cosmos to consciousness. PhilArchive.
<https://philarchive.org/rec/CARIE>

Fuchs, C. A., Mermin, N. D., & Schack, R. (2014). An introduction to QBism with an application to the locality of quantum mechanics. *American Journal of Physics*, 82(8), 749–754.
<https://doi.org/10.1119/1.4874855>

Kagan, B. J., Kitchen, A. C., Tran, N. T., Habibollahi, F., Khajehnejad, M., Parker, B. J., ... & Friston, K. J. (2022). In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. *Neuron*, 110(23), 3952–3969.e8.
<https://doi.org/10.1016/j.neuron.2022.09.001>

Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183–191.
<https://doi.org/10.1147/rd.53.0183>

Rovelli, C. (1996). Relational quantum mechanics. *International Journal of Theoretical Physics*, 35(8), 1637–1678.
<https://doi.org/10.1007/BF02302261>

Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: From consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450–461.
<https://doi.org/10.1038/nrn.2016.44>

Vazza, F., & Feletti, A. (2020). The quantitative comparison between the neural network and the cosmic web. *Frontiers in Physics*, 8, 525731.
<https://doi.org/10.3389/fphy.2020.525731>

Vopson, M. M. (2025). Is gravity evidence of a computational universe? *AIP Advances*, 15(4), 045035. <https://doi.org/10.1063/5.0264945>

Wheeler, J. A. (1990). Information, physics, quantum: The search for links. In W. Zurek (Ed.), *Complexity, entropy, and the physics of information* (pp. 3–28). Addison-Wesley.

Wigner, E. P. (1960). The unreasonable effectiveness of mathematics in the natural sciences. *Communications on Pure and Applied Mathematics*, 13(1), 1–14.
<https://doi.org/10.1002/cpa.3160130102>

Zeilinger, A. (1999). A foundational principle for quantum mechanics. *Foundations of Physics*, 29(4), 631–643.
<https://doi.org/10.1023/A:1018820410908>

